HW3

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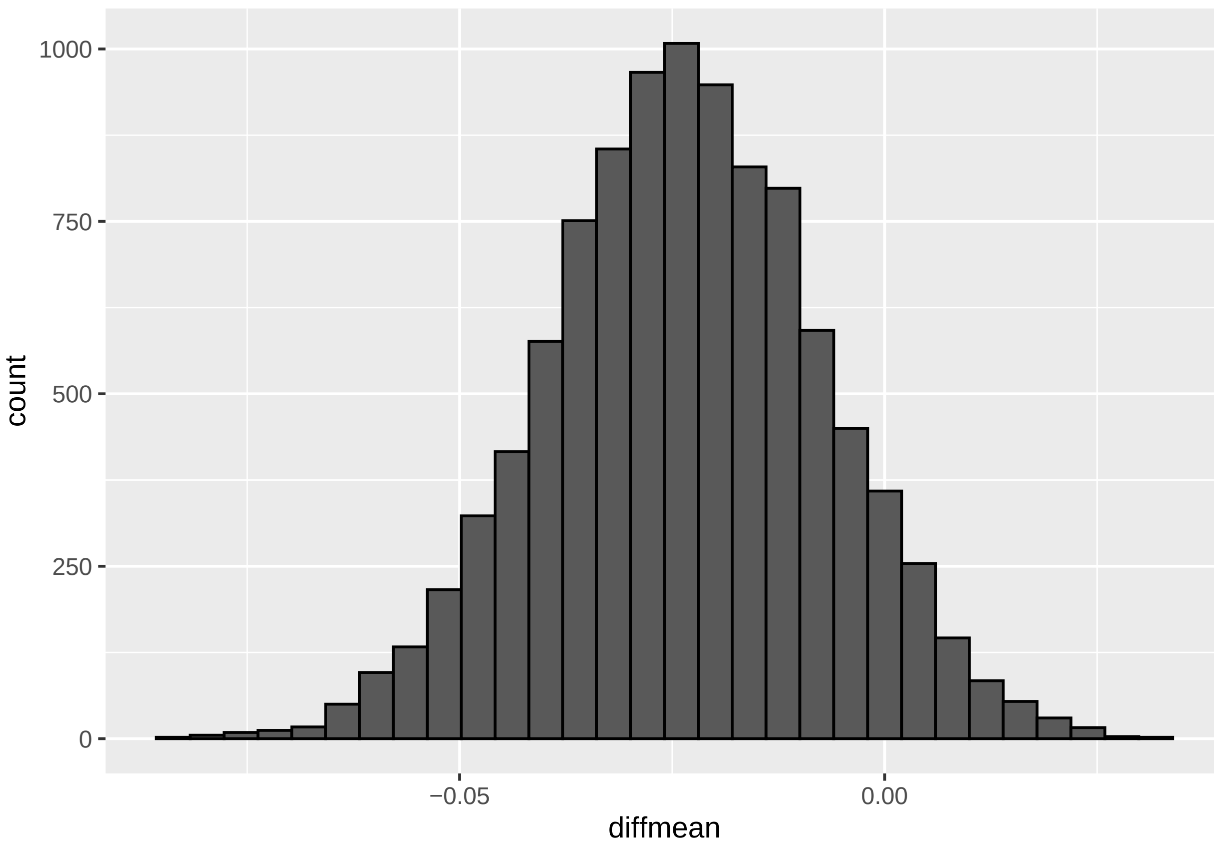
GitHub Link: https://github.com/brianliangg/SDS315_HW3

Problem 1

Theory A



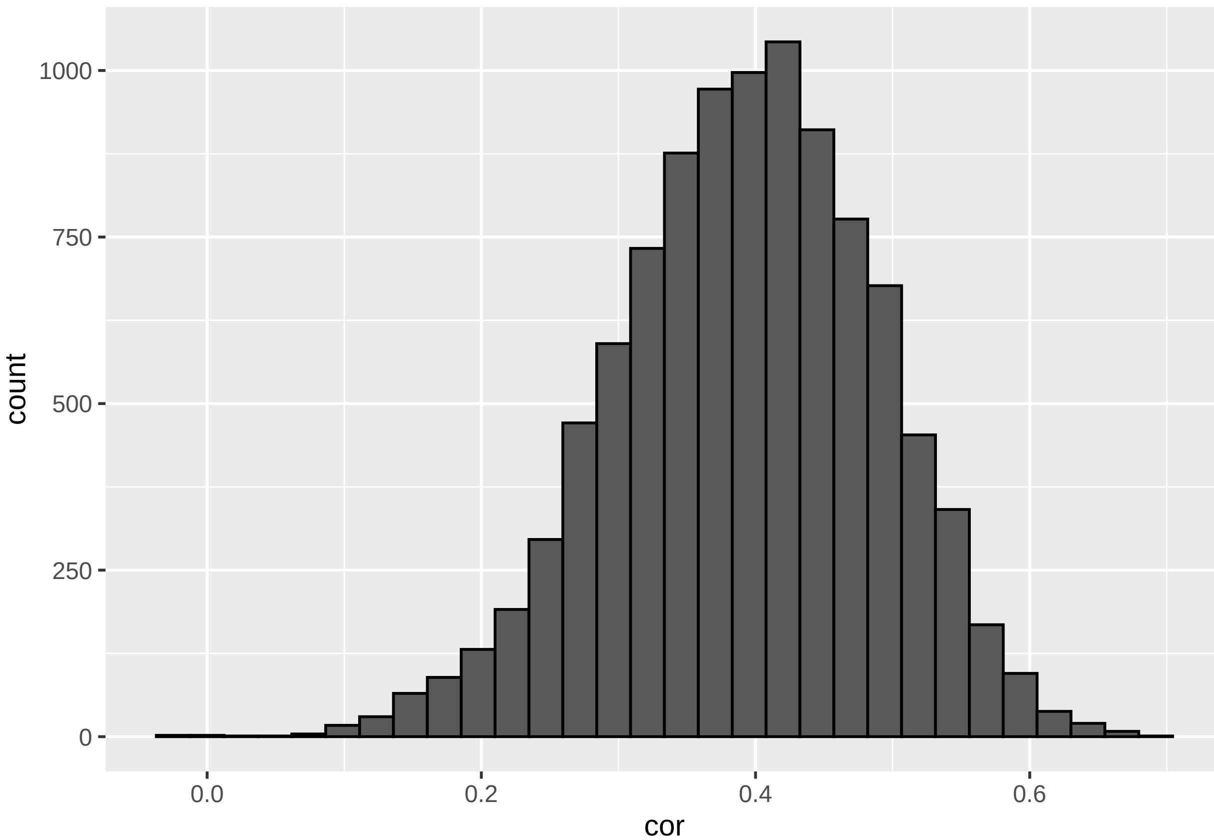

```
##       name       lower       upper level     method    estimate
## 1 diffmean -0.05588229 0.008218527  0.95 percentile -0.01669025
```

Claim: The theory that was claimed stated that, "Gas stations charge more if they lack direct competition in sight."

Evidence: This theory, however, is unsupported as from both the bootstrap sampling distribution histogram and confidence interval that was created as 0 is included in the range. A 0 signifies that there is no difference between gas prices if there is no direct competition in sight and if there is.

Conclusion: This means that for a 95% confidence interval, the difference found is not statistically significant at the 5% level as it contains 0. This means that Theory A is not supported by the data.

Theory B



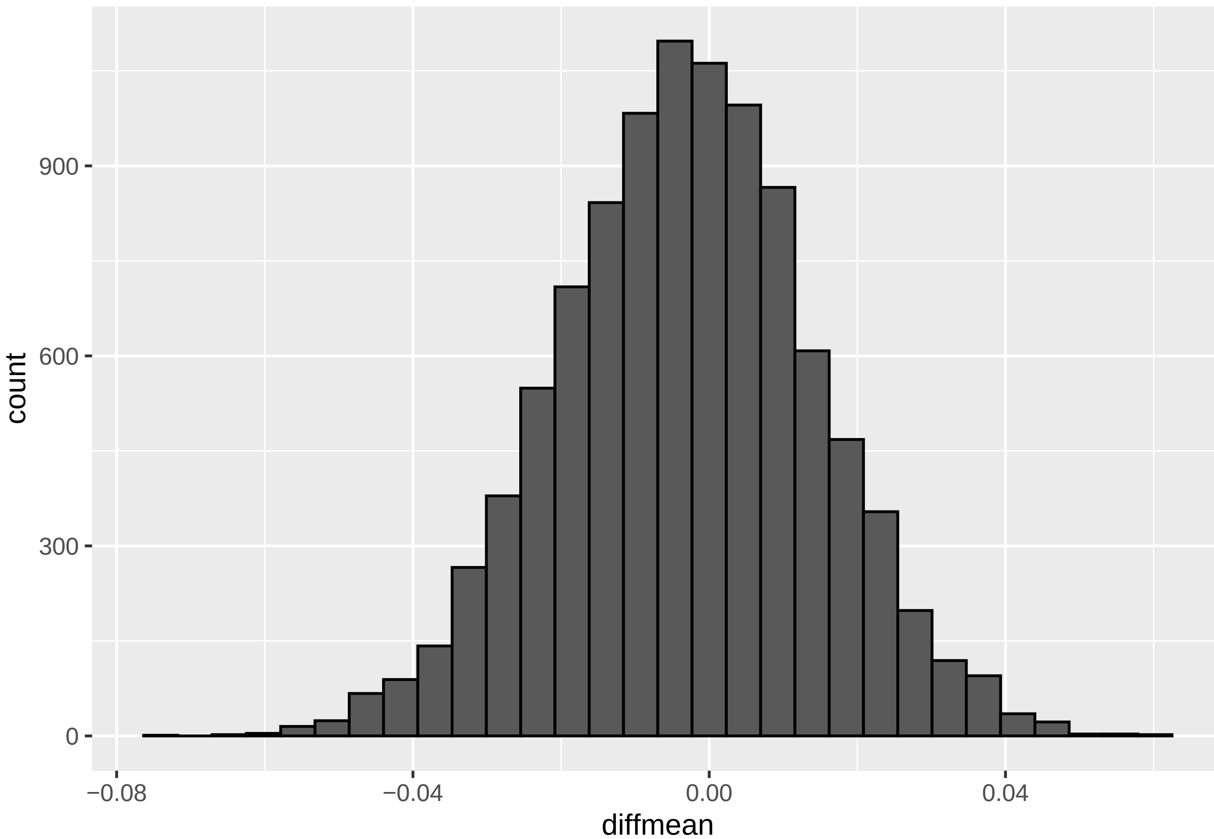

```
##   name     lower     upper level     method  estimate
## 1  cor 0.1927011 0.5652933  0.95 percentile 0.5209655
```

Claim: Theory B claims that, "The richer the area, the higher the gas prices."

Evidence: The evidence shows that from the bootstrap sampling distribution of the correlation coefficient between gas prices and median household income in that area, the theory is supported by the data. The histogram and the confidence interval that were found for this distribution demonstrate that this correlation is statistically significant at the 5% level as 0, which signifies no correlation, is not included in the interval.

Conclusion: With 95% confidence, there is an association between gas prices and median household income in that area that lies in the range from a weak correlation to a moderate correlation, meaning Theory B is supported by the data.

Theory C



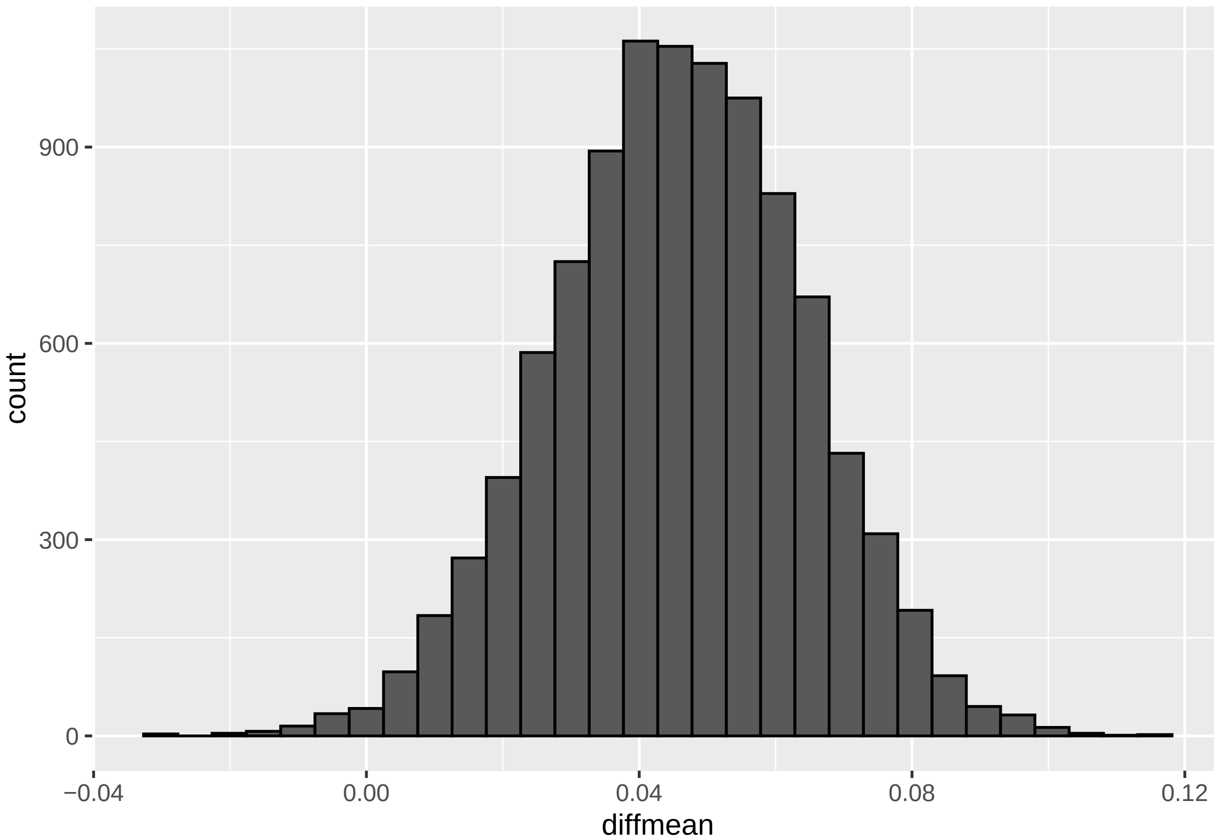

```
##       name       lower      upper level     method   estimate
## 1 diffmean -0.03797602 0.03093135  0.95 percentile 0.02086587
```

Claim: Theory C states that, “Gas stations at stoplights charge more.”

Evidence: This theory is unsupported by the data as 0 is included in the range of possible values for the difference in average gas prices by if that gas station has a stoplight. A 0 signifies that there is no difference between the gas prices if there is a stoplight or if there is not a stoplight.

Conclusion: This means that for a 95% confidence interval, the difference found is not statistically significant at the 5% level as it contains 0, meaning Theory C is not supported by the data.

Theory D



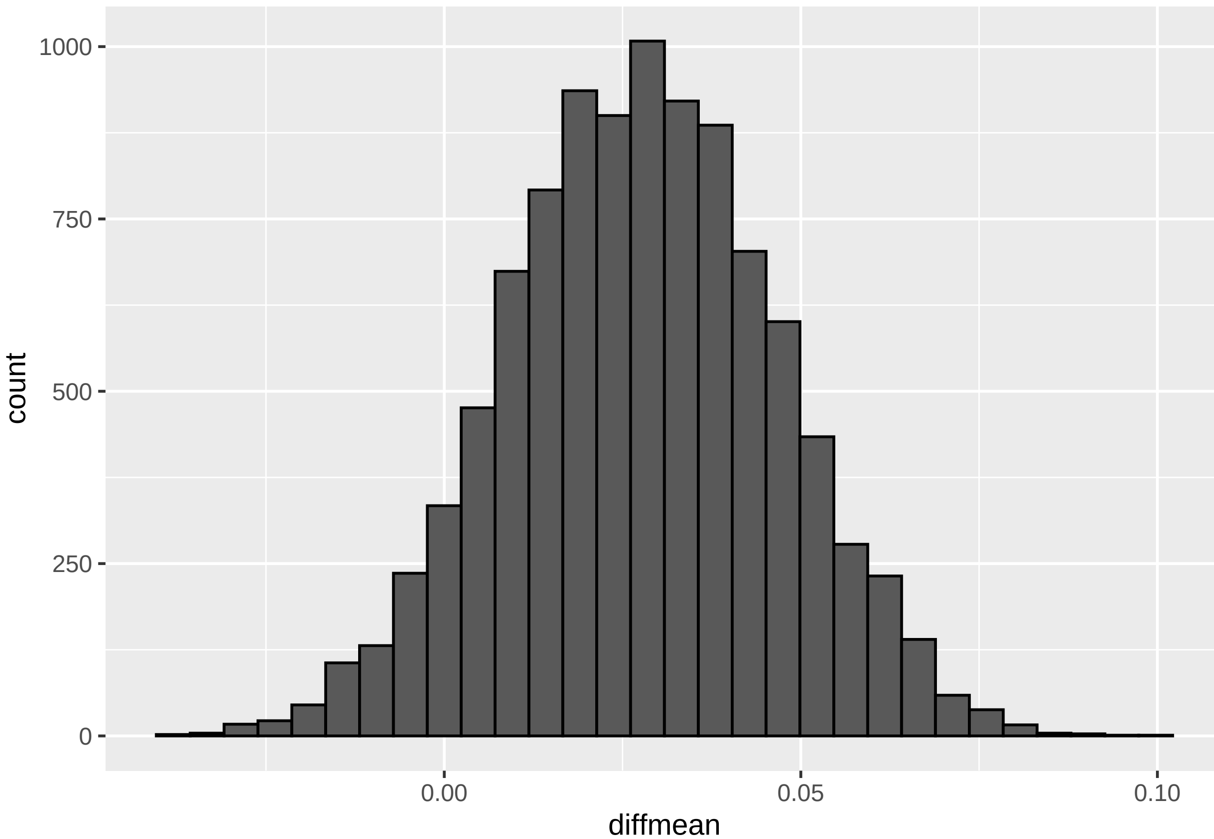

```
##       name       lower     upper level     method   estimate
## 1 diffmean 0.009159226 0.0810353  0.95 percentile 0.03240196
```

Claim: Theory D claims that, "Gas stations with direct highway access charge more."

Evidence: This theory is supported by the data as the histogram and confidence interval of the difference between the average gas price by if the gas station has direct highway access does not include 0. A 0 would signify that there is not difference between the prices based on highway access.

Conclusion: The difference in price between gas stations that have direct highway access is about $0.01 to $0.08, with 95% confidence. This signifies that Theory D is supported by the data.

Theory E




```
##       name       lower      upper level     method   estimate
## 1 diffmean -0.00979466 0.06445678  0.95 percentile 0.03101159
```

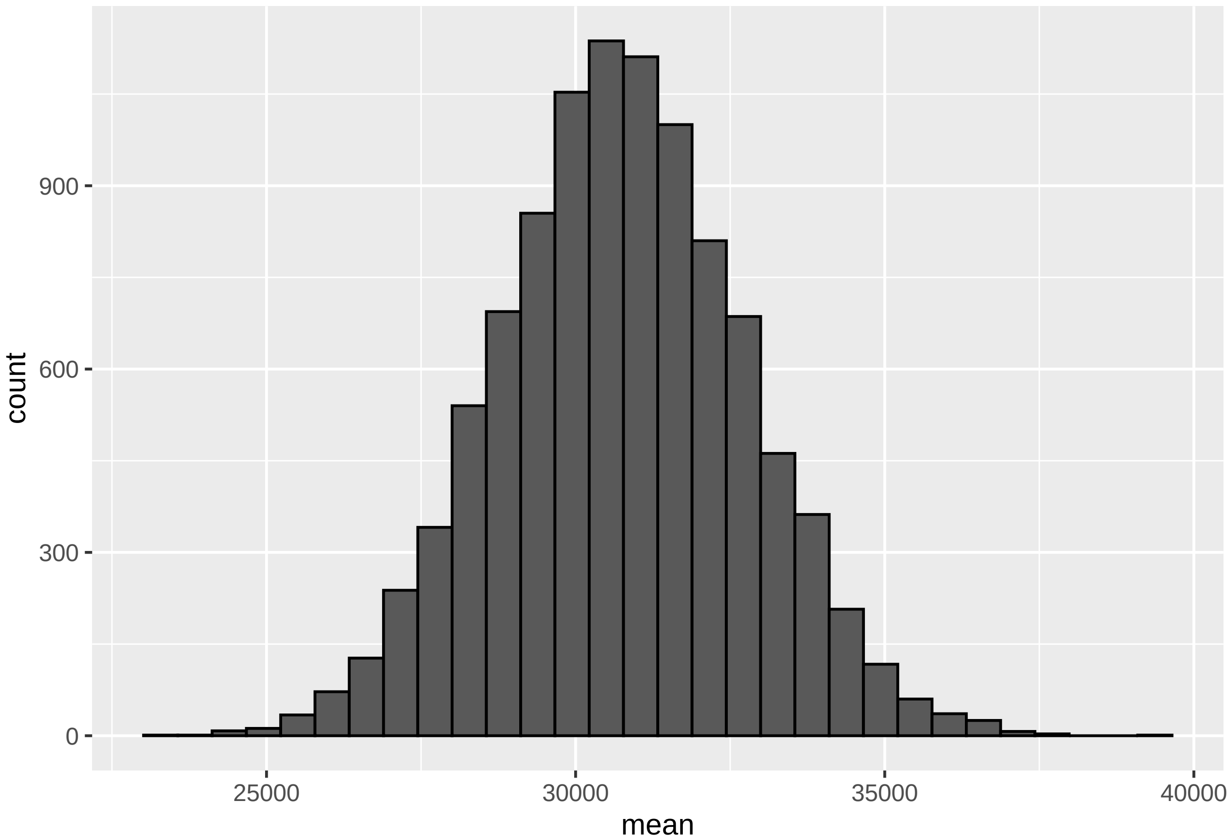
Claim: Theory E claims that "Shell charges more than all other non-Shell brands."

Evidence: This theory is not supported by the data as 0 is included in the range of the difference of average gas prices from Shell versus non-Shell brands. The histogram and confidence interval for this difference of means have 0 as a possible value which means there would be no difference between the prices Shell charges in comparison to the other brands.

Conclusion: Theory E is not supported by the data as with 95% confidence, the difference is not statistically significant at the 5% level as it contains 0 in the range of possible values.

Problem 2

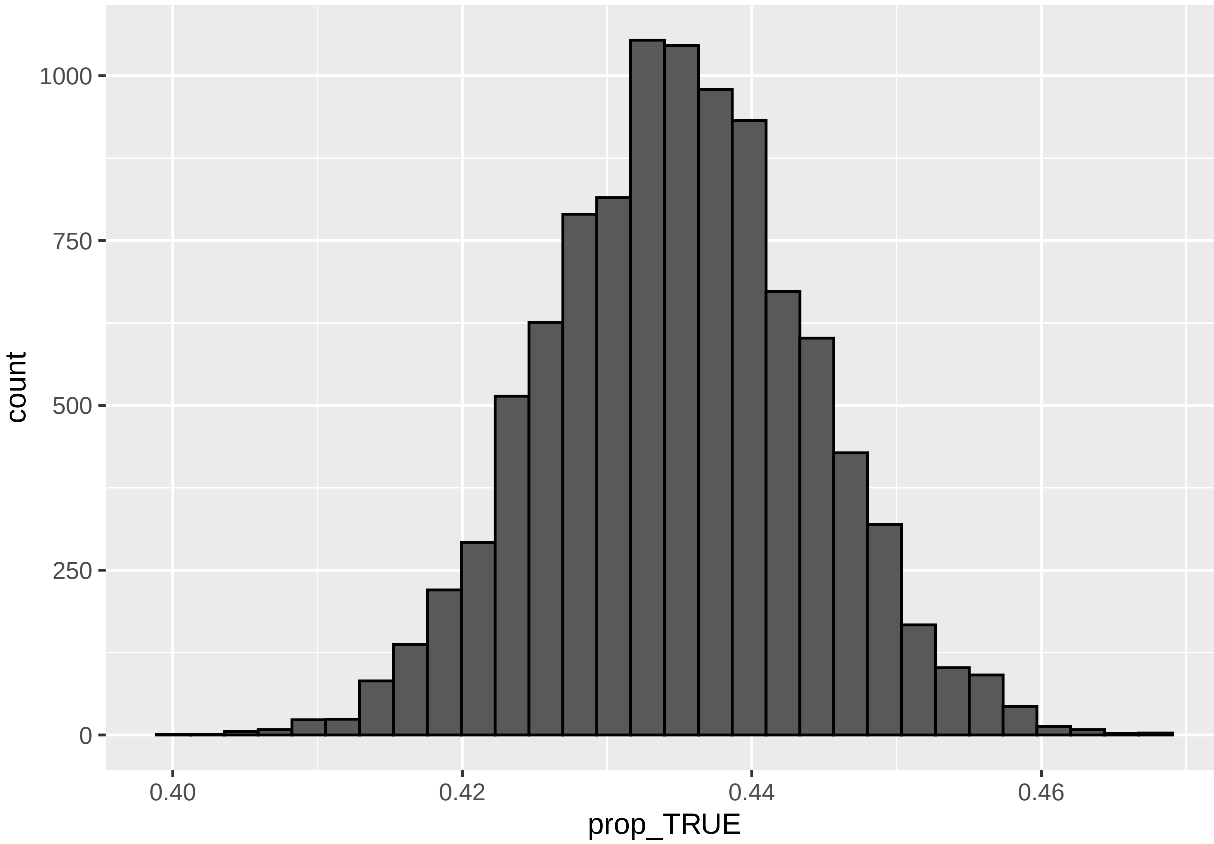
Part A




```
##   name    lower    upper level     method estimate
## 1 mean 26885.67 34637.48  0.95 percentile 31819.31
```

With 95% confidence the average mileage of 2011 S-Class 63 AMGs that were on the used car market on car.com was somewhere between 26,885.67 to 34,637.48 miles.

Part B

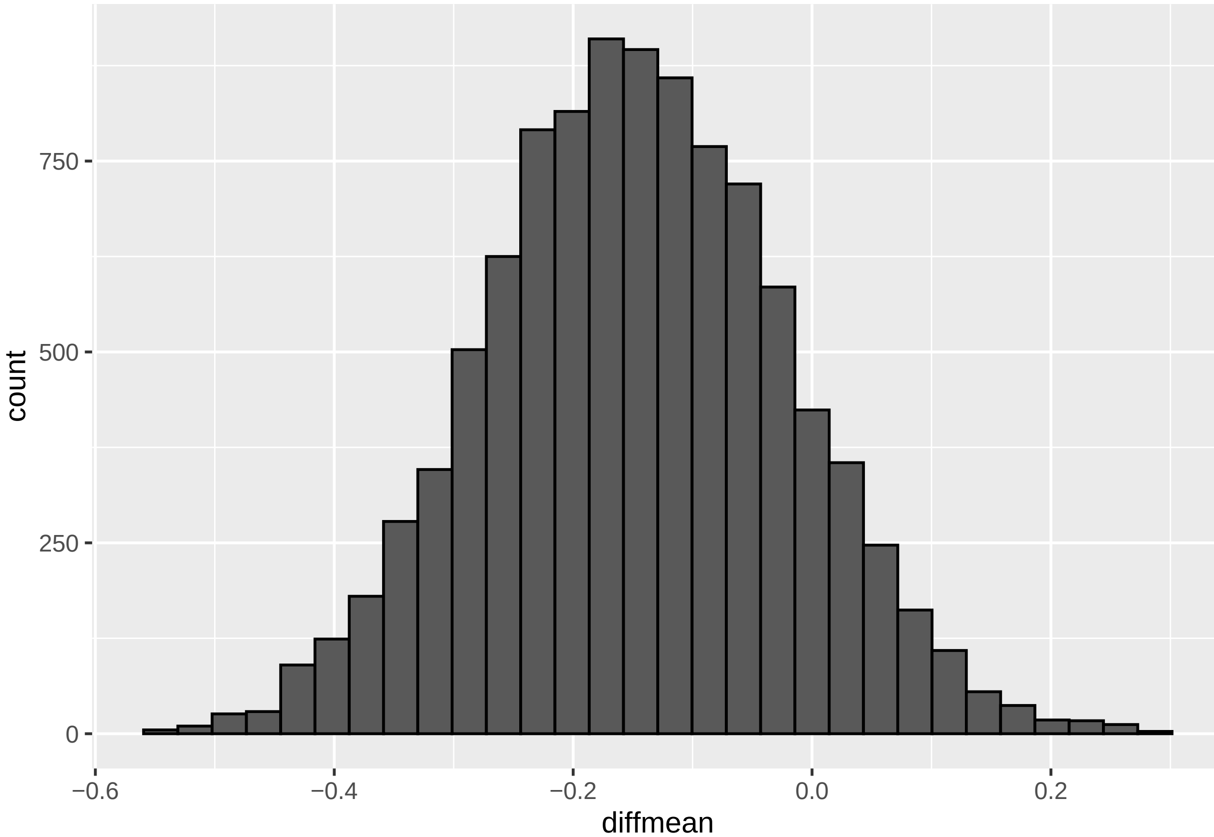



```
##        name     lower     upper level     method  estimate
## 1 prop_TRUE 0.4170993 0.4527518  0.95 percentile 0.4319834
```

With 95% confidence, the percentage of all 2014 S-Class 550s that were painted black was somewhere between 41.71% to 45.28% of the ones sold on cars.com.

Problem 3

Part A




```
##       name      lower     upper level     method  estimate
## 1 diffmean -0.3916786 0.1005722  0.95 percentile 0.0106383
```

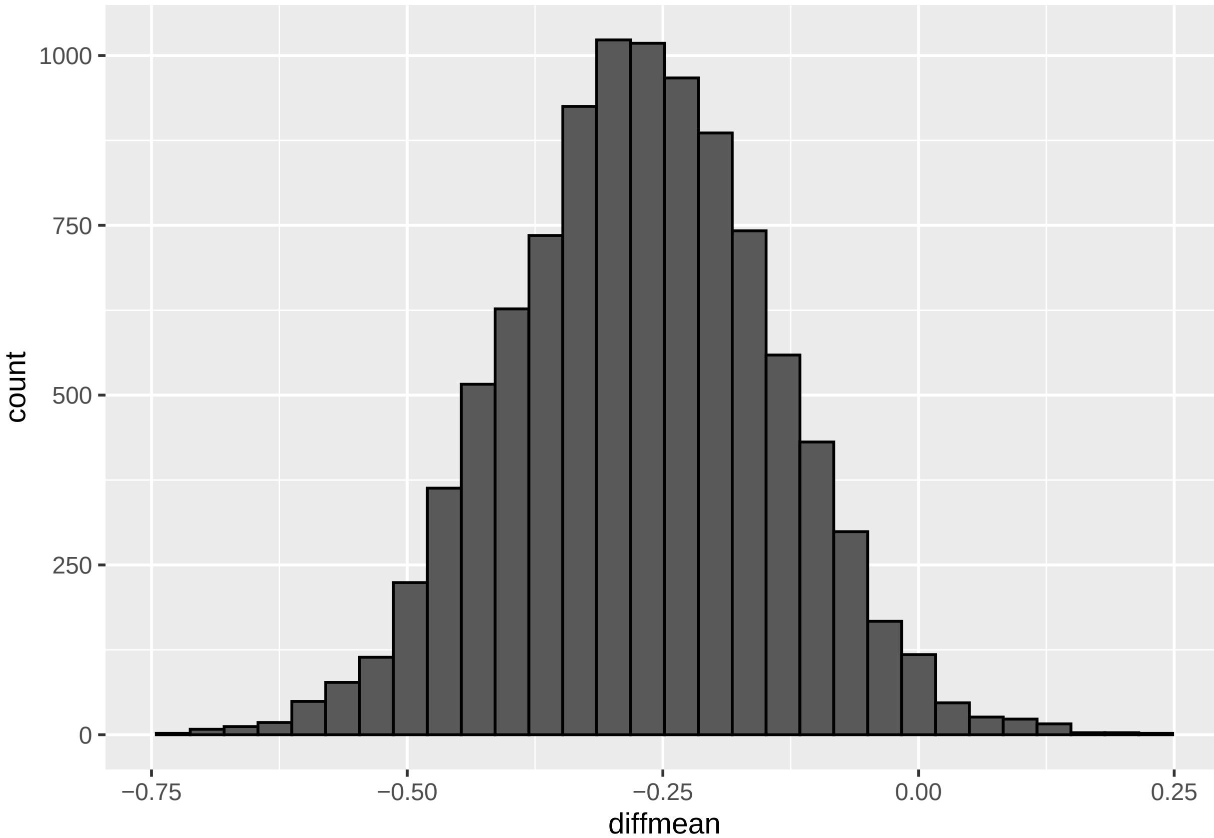
Question: The question I am trying to answer is of the shows "Living with Ed" and "My Name is Earl," which of these make people happier?

Approach: My approach was to create a bootstrap sampling distribution of the difference in the average happiness rating of both shows. I then constructed a 95% confidence interval based on the bootstrap sampling distribution to determine the results. In addition, I used a histogram to visualize this bootstrap sampling distribution.

Results: With 95% confidence, the difference in the average happiness rating received by the shows "Living with Ed" and "My Name is Earl," lies somewhere between -0.392 and 0.097. However, 0 is included with in this range of values.

Conclusion: Based on the confidence interval and histogram of the difference in the average happiness ratings of the shows "Living with Ed" and "My Name is Earl," it suggests that there is no difference as 0 is contained in the range of values. This 0 signifies that there is likely no difference in the ratings.

Part B




```
##       name      lower       upper level     method   estimate
## 1 diffmean -0.5212375 -0.02036616  0.95 percentile -0.2046855
```

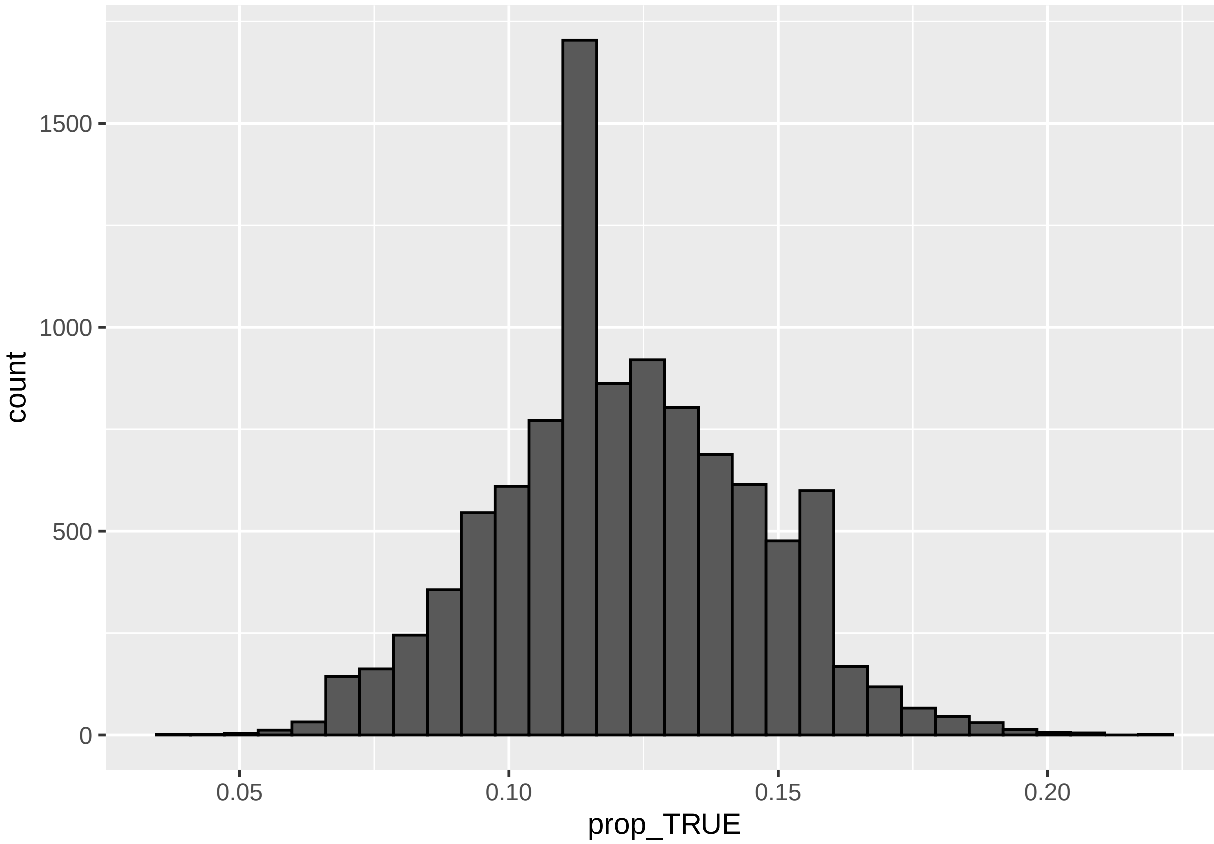
Question: The question I am trying to answer is of the shows “The Biggest Loser” and “The Apprentice: Los Angeles,” which reality/contest show made people more annoyed?

Approach: My approach was to create a bootstrap sampling distribution of the difference in the average annoyance rating of both shows. I then constructed a 95% confidence interval based on the bootstrap sampling distribution to determine the results. In addition, I used a histogram to visualize this bootstrap sampling distribution.

Results: With 95% confidence, the difference in the average annoyance rating received by the shows “The Biggest Loser” and “The Apprentice: Los Angeles,” lies somewhere between -0.5212 and -0.0203. This is statistically significant as 0 is not contained within that range of values at the 5% level.

Conclusion: Based on the confidence interval and histogram the annoyance ratings of the shows “The Biggest Loser” and “The Apprentice: Los Angeles” suggests that “The Biggest Loser” annoyed viewers more as it had a higher mean rating for annoyance that was somewhere between 0.0203 and 0.5212 of a rating higher than “The Apprentice: Los Angeles.”

Part C




```
##        name      lower     upper level     method   estimate
## 1 prop_TRUE 0.07734807 0.1712707  0.95 percentile 0.09392265
```

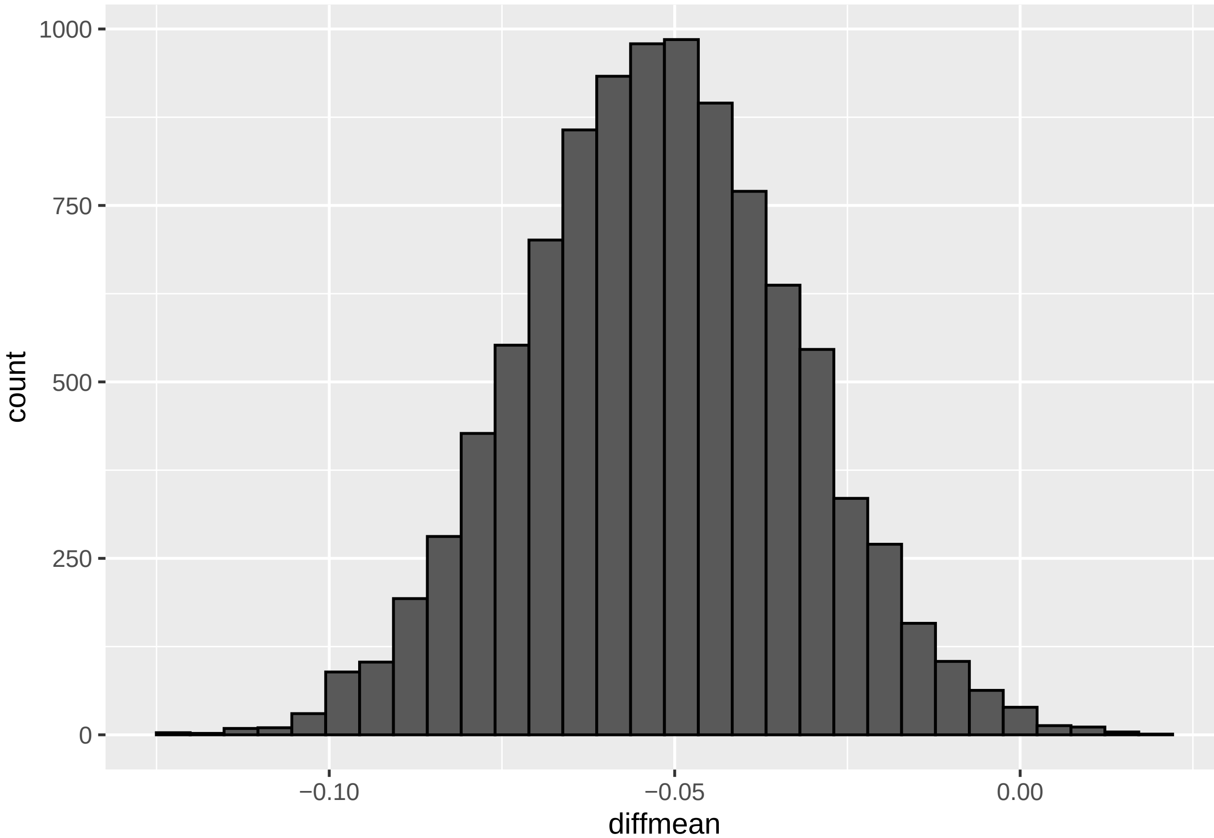
Question: The question I am trying to answer is for the show "Dancing with the Stars," what proportion of American TV watchers would we expect to give a response of 4 or greater to the "Q2_Confusing" question?

Approach: My approach was to create a bootstrap sampling distribution of the proportion of ratings were a 4 or higher for the question on if viewers were confused. I then constructed a 95% confidence interval based on the bootstrap sampling distribution to determine the results. In addition, I used a histogram to visualize this bootstrap sampling distribution.

Results: With 95% confidence, the proportion of American TV watchers we would expect to give a response of 4 or greater to the "Q2_Confusing" question lies somewhere between 0.0773 and 0.1713.

Conclusion: Based on the confidence interval and histogram, the percentage of American TV watchers we would expect to give a response of 4 or greater to the "Q2_Confusing" question is somewhere between 7.73% to 17.13%.

Problem 4




```
##       name       lower       upper level     method    estimate
## 1 diffmean -0.09059633 -0.01297125  0.95 percentile -0.04123124
```

Question: The question I am trying to answer is "Whether instead the data favors the idea that paid search advertising on Google creates extra revenue for EBay?"

Approach: My approach was to first create a new variable that would have the ratio of revenue from before to after when paid search advertising was stopped. I then created a bootstrap sampling distribution of the difference in the average revenue ratio by if it was the treatment or control group. I then constructed a 95% confidence interval based on the bootstrap sampling distribution to determine the results. In addition, I used a histogram to visualize this bootstrap sampling distribution.

Results: With 95% confidence, the difference in average revenue ratio for DMAs in treatment group versus the control group lies somewhere between -0.091 and -0.013. This is statistically significant at the 5% level as 0 is not contained within the range of the 95% confidence interval.

Conclusion: Based on the confidence interval and histogram the data favors the idea that paid search advertising on Google does create extra revenue for Ebay. This is because the difference between the average revenue ratio between DMAs who did not have their adwords paused to those who did was somewhere between 0.013 to 0.091.